Shoshan

A Hebrew lemmatizer that never invents a word.

Lemmatization is the quiet workhorse of Hebrew search: it collapses a word's many inflected forms to one dictionary lemma, so a query for one form finds them all. Run blindly over millions of tokens, its mistakes are never reviewed — which makes one kind of mistake unacceptable. A lemmatizer that returns a **plausible wrong word** silently merges unrelated terms or makes a word disappear from the index, and no one notices. Shoshan is built so that this cannot happen.

Same sentences, two systems

Below, real Hebrew sentences run through Shoshan and through DictaBERT-lex, the strongest Hebrew lemmatizer. ● blue = a real lemma retrieved from a fixed bank; ● orange = generated. Shoshan retrieves or makes a bounded edit of the word; DictaBERT-lex predicts a single token of its vocabulary, so it can land on an unrelated word or on `[BLANK]` (shaded).

DictaBERT-lex	Shoshan	מילה
הוא	הוא	הוא
ניסה	ניסה	ניסה
חיפש	מצא	למצוא
את	את	את
מפתח	מפתח	המפתחות
אבוד	אבוד	האבודים
מתחת	מתחת	מתחת
מושב	מושב	למושב
נהג	נהג	הנהג

DictaBERT-lex reads **למצוא** (“to find”) as **חיפש** (“searched”) — a fluent, unrelated word. Shoshan retrieves the true lemma **מצא**.

DictaBERT-lex	Shoshan	מילה
שופט	שופט	השופט
קבע	קבע	קבע
כי	כי	כי
ביקש	מבקש	המבקש
לא	לא	לא
הוכיח	הוכיח	הוכיח
את	את	את
טענה	טענה	טענתו

In a legal sentence, **המבקש** (“the petitioner”) drifts to **ביקש** (“asked”). Filed under the wrong key, that document vanishes from a search for “petitioner.”

DictaBERT-lex	Shoshan	מילה
[BLANK]	ג'דיי	הג'דיי
שלף	שלף	שלף
את	את	את
[BLANK]	לייטסייבר	הלייטסייבר
זוהר	זוהר	הזוהר
של	של	שלו
לפני	לפני	לפני
קרב	קרב	הקרב

The loanword **הלייטסייבר** is no single vocabulary token, so DictaBERT-lex emits `[BLANK]` — the word drops out of the index. Shoshan strips the clitic and returns **לייטסייבר**.

DictaBERT-lex	Shoshan	מילה
משורר	משורר	המשורר
כתב	כתב	כתב
על	על	על
אהבה	אהבה	אהבתו
גדול	גדול	הגדולה
ספר	ספר	בספרו
אחרון	אחרון	האחרון

On ordinary text the two agree completely. The gap is not noise — it is concentrated exactly where the words are unfamiliar, which is where a corpus project needs the most help.

Why Shoshan can't hallucinate

> Every Shoshan output is either an entry from a fixed bank of real Hebrew lemmas, or a **bounded edit of the input word** — strip a prefix, fix a suffix. There is no path by which it can emit a word unrelated to the one on the page. DictaBERT-lex instead predicts the lemma as the single most likely token of its ~128K-word-piece vocabulary; when the right lemma is a fluent neighbour, or is not in that vocabulary at all, the output is wrong or empty.

Shoshan is not perfect: on a rare unseen verb it can return a clipped but still word-shaped stem (e.g. **צפ** → **צפו** instead of **צפה**). The guarantee is narrower, and exactly the one that matters for retrieval — it can be wrong, but it can never return a word the input could not have produced, and it never returns nothing.

The numbers

0.0%
hallucinated lemmas on unseen words
(DictaBERT-lex: 12.3%)

0.959
B³ consistency F_1, out of domain
(DictaBERT-lex: 0.919)

92.4%
Shoshan lemma accuracy, held-out domains
(in-domain 94.3% — both Shoshan)

Both accuracy figures are Shoshan's own — out-of-domain and in-domain. We compare the two systems on B³ rather than exact match, because their lemma conventions differ and exact-match accuracy is not comparable across them. Trained on only the openly redistributable 38% of the IAHLT treebank, Shoshan matches a baseline trained on far more data on that consistency, while removing its hallucinations outright.

How it works

1 · Retrieve
Embed the word in its sentence; pull the nearest lemma from a bank of ~118K real Hebrew lemmas.

2 · Check
A coverage gate asks whether that lemma's letters actually sit inside the word. If not, the word is likely unknown.

3 · Transduce
For unknown words, build the lemma by editing the word itself — a bounded operation that stays a real form.

Extending Shoshan to a new domain is adding rows to the bank and re-encoding them — no retraining. The model even flags its own unknown words as a ranked list to curate.

Get it. code github.com/ivrit/shoshan · weights huggingface.co/noamor/shoshan · data noamor/shoshan-data · live demo noamor/shoshan-demo

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Shoshan is fine-tuned from DictaBERT. Comparison figures are on identical aligned tokens; DictaBERT-lex was trained on substantially more data, including the registers held out here. Examples are real model outputs, selected to illustrate the failure modes.